

Juwon Kim

Postdoctoral Researcher, School of Electrical Engineering, KAIST
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Research Interests

Kernel, Filesystem, Flash storage, Device firmware (Flash Translation Layers), and Garbage collection.

Education

Ph.D. in Electrical Engineering Mar. 2023 – Aug. 2026
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea

- Advisor: Prof. Youjip Won

M.S. in Electrical Engineering Mar. 2021 – Feb. 2023
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea

- Thesis: *IPLFS: Log-Structured Filesystem without Garbage Collection*
- Advisor: Prof. Youjip Won

B.S. in Electrical Engineering Mar. 2016 – Feb. 2021
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea

- Individually Designed Major in Computer Science and Physics

Publications

- [1] **[ACM SOSP '26]** Juwon Kim, Hwanseok Yoo, Seung Won Yoo, and Youjip Won, “GAZE: Rearchitecting Log-Structured Filesystem for Garbage Collection on Zoned Namespace Flash Storage”, accepted for publication in *Proc. of ACM Symposium on Operating Systems Principles (SOSP)* (acceptance rate: 15%), Prague, Czechia, Sep.–Oct. 2026.
- [2] **[IEEE NVMSA '26]** Jihoon Seok, Juwon Kim, Youjip Won, “LeaST: Learned Sector Translation for Skipped Physical Page Address”, in *Proc. of IEEE Non-Volatile Memory Systems and Applications Symposium (NVMSA)*, Aug. 2026.
- [3] **[USENIX FAST '25]** Juwon Kim, Seungjae Lee, Joontaek Oh, Dongkun Shin, Youjip Won, “D2FS: Device-Driven Filesystem Garbage Collection”, in *Proc. of USENIX Conference on File and Storage Technologies (FAST)* (acceptance rate: 21%), Santa Clara, USA, Feb. 2025.
- [4] **[USENIX FAST '25]** Jieun Kim, Joontaek Oh, Juwon Kim, Seung Won Yoo, Youjip Won, “OPIMQ: Order Preserving IO Stack for Multi-Queue Block Device”, in *Proc. of USENIX Conference on File and Storage Technologies (FAST)* (acceptance rate: 21%), Santa Clara, USA, Feb. 2025.
- [5] **[ACM SAC '25]** Juwon Kim, Dongeon Kim, Seung Won Yoo, Myoengin Cheon, Joontaek Oh, Youjip Won, “lwFSCK: Light-weight Filesystem Check”, in *Proc. of ACM Symposium on Applied Computing (SAC)*, Catania, Italy, Mar. 2025.
- [6] **[USENIX ATC '22]** Juwon Kim, Minsu Jang, Danish Muhammad Teeshen, Joontaek Oh, Youjip Won, “IPLFS: Log-Structured File System without Garbage Collection”, in *Proc. of USENIX Annual Technical Conference (ATC)* (acceptance rate: 16%), San Diego, USA, Jul. 2022.

Posters and Workshops

- [1] **[ACM HotStorage '26]** Hwanseok Yoo, Juwon Kim, Youjip Won, “Manycore Scalable Logging for Log-Structured Filesystem,” in *Proc. of ACM Workshop on Hot Topics in Storage and*

File Systems (HotStorage), Prague, Czechia, Sep. 2026.

- [2] **[ACM SOSP '25] Juwon Kim**, Hwanseok Yoo, Youjip Won, “Accelerating Filesystem Garbage Collection for Zoned Flash Storage,” in *Proc. of ACM Symposium on Operating Systems Principles (SOSP)*, Seoul, Republic of Korea, Oct. 2025 (**Poster**).
- [3] **[ACM SOSP '25] Hwanseok Yoo, Juwon Kim**, Youjip Won, “Manycore Scalability of Modern Log-Structured Filesystem: Revisited,” in *Proc. of ACM Symposium on Operating Systems Principles (SOSP)*, Seoul, Republic of Korea, Oct. 2025 (**Poster**).
- [4] **[USENIX FAST '22] Juwon Kim**, Minsu Jang, Danish Muhammad Teeshen, Joontaek Oh, Youjip Won, “IPLFS: Log-Structured File System without Garbage Collection,” in *Proc. of USENIX Conference on File and Storage Technologies (FAST)*, Santa Clara, USA, Feb. 2022 (**Poster**).

Patents

- [1] Youjip Won and **Juwon Kim**, “Method and Device for Managing Host Area,” Korean Patent Application No. 10-2025-0174419, filed Nov. 2025.
- [2] **Juwon Kim** and Youjip Won, “Storage Device Performing Garbage Collection,” Korean Patent Application No. 10-2025-0090393, filed Jul. 2025. (*Joint filing: SK Hynix & KAIST*)

Awards & Honors

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| • Student Scholarship, ACM SOSP, Czechia | Sep. 2026 |
| • Student Grant (Full), USENIX FAST, USA | Feb. 2025 |
| • NVRAMOS 2023 OpenSSD Best Practice Award, Gold Prize, KIISE, Korea | Nov. 2023 |
| • Student Grant (Full), USENIX FAST, USA | Feb. 2022 |

Research Experience

Postdoctoral Researcher Operating Systems Laboratory, KAIST	Sep. 2026 – Present Daejeon, Korea
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Visiting Researcher SK Hynix	Jul. 2025 – Aug. 2025 Bundang, Korea
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Filesystem Garbage Collection for Zoned Mobile Storage

- Investigated filesystem garbage collection overhead on mobile platforms and explored host-side optimizations for zoned UFS devices.

Zoned UFS, F2FS, Garbage collection, Mobile storage, Linux kernel

Graduate Researcher (Ph.D. + M.S.) Operating Systems Laboratory, KAIST	Mar. 2021 – Aug. 2026 Daejeon, Korea
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Log-Structured Filesystem for Large-Zone Reclamation

- Rearchitected a log-structured filesystem for efficient reclamation of large zones in Zoned Namespace flash storage, eliminating fundamental CPU and I/O bottlenecks in conventional filesystem garbage collection.

F2FS, Zoned Namespace SSD, Garbage collection, Linux kernel, Memory management

Device-Driven Filesystem Garbage Collection

- Designed a host–device cooperative filesystem architecture that delegates filesystem garbage collection to the storage device, reducing host CPU utilization and host–device data movement.

Filesystem, Flash translation layer, Garbage collection, Host–device cooperation, NVMe

Log-Structured Filesystem without Garbage Collection

- Designed a log-structured filesystem that eliminates filesystem-level garbage collection by virtually expanding the filesystem partition over additional physical storage space.

F2FS, Log-structured filesystem, Garbage collection, Flash storage, Linux kernel

Undergraduate Research Intern
Operating Systems Laboratory, KAIST

Jan. 2020 – Feb. 2021
Daejeon, Korea

Log-Structured Filesystem Design

- Investigated log-structured filesystem design and filesystem garbage collection mechanisms under Prof. Youjip Won.

Filesystem, F2FS, Linux kernel, Flash storage

Work Experience

Machine Learning Engineer Intern
Penta Security Systems Inc.

Jun. 2018 – Jan. 2019
Seoul, Korea

CAN Data-Based Anomaly Detection for Electric Vehicles

- Designed and implemented a GRU-D autoencoder for detecting anomalous vehicle behavior from CAN bus data.

Machine learning, GRU-D, Autoencoder, CAN bus, Anomaly detection ““

Research Projects

Ultra-Low-Latency Storage-Driven I/O Subsystem for Large Language Models *Jul. 2026 – Present*

InnoCORE, Ministry of Science and ICT, Korea

Order Preserving Distributed Storage System
NRF, Korea

Jun. 2025 – Present

Optimizing Filesystem Garbage Collection for Zoned Flash Storage *Sep. 2024 – Present*
SNU-SKH Solution Research Center, SK Hynix, Korea

Device-Driven Filesystem Garbage Collection
SNU-SKH Solution Research Center, SK Hynix, Korea

Sep. 2023 – Aug. 2024

Collection and Integrated Analysis of Automotive Inter/Intra System Artifacts *Jan. 2024 – Feb. 2025*
IITP, Korea

Future Scalable OS: Log-Structured Filesystem without Garbage Collection *Jan. 2021 – May 2023*
IITP, Korea

Teaching Experience

- Head TA, Operating Systems and System Programming for EE (EE415), KAIST *Sep.–Dec. 2025*
- TA, Special Topics in Computer Engineering (EE817), KAIST *Feb.–Jun. 2025*
- TA, Operating Systems and System Programming for EE (EE415), KAIST *Sep.–Dec. 2024*
- Head TA, Programming Structure for EE (EE209), KAIST *Feb.–Jun. 2024*
- TA, Programming Structure for EE (EE209), KAIST *Sep.–Dec. 2023*
- TA, Programming Structure for EE (EE209), KAIST *Feb.–Jun. 2023*

Service

- Artifact Evaluation Committee, USENIX FAST '27 Spring Cycle *Jun. 2026 – Jul. 2026*

- Lab Manager, Operating Systems Laboratory, KAIST
- Program Chair Assistant, USENIX FAST '24

Jan. 2026–Jul. 2026

Sep. 2023–Sep. 2024

Technical Skills

Systems: F2FS, flash translation layer, NVMe driver, block I/O layer, ZNS, Page Management Subsystem

Languages: C, Python, C++, SQL, Verilog

Tools: git, vim, gnuplot, ctags, cscope, ftrace, blktrace